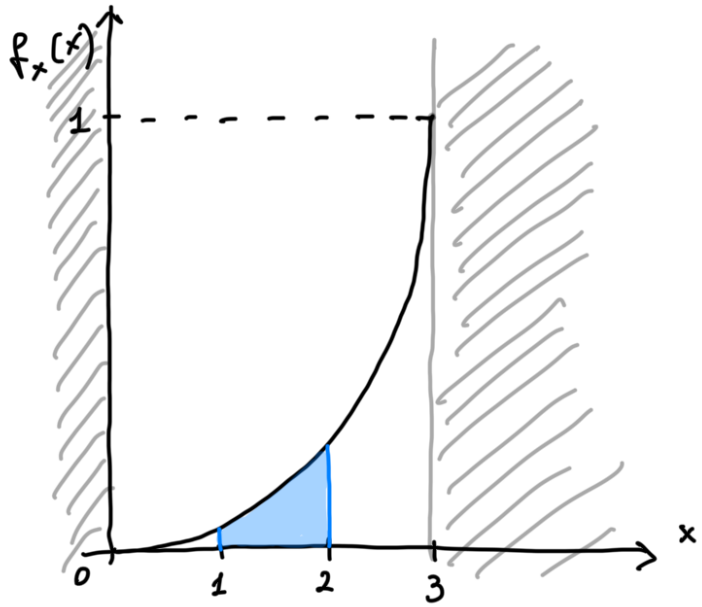
Esercizio 4. Sia X una variabile aleatoria continua con densità $f_X(x) = c\,x^2$ per $0 \leq x \leq 3$, e $f_X(x) = 0$ altrimenti.

a) Determinare c.

b) Calcolare $\mathrm{P}(1 \leq X \leq 2)$ e $\mathrm{P}(X = 1.5)$.

c) Determinare la funzione di ripartizione F_X.

d) Calcolare $\mathrm{E}[X]$ e $\mathrm{var}(X)$.



$f_X(x) = cx^2$ per $0 \le x \le 3$

a) $\int_0^3 cx^2\,dx = 1 \quad \Rightarrow \quad c\,\frac{x^3}{3}\Big|_0^3 = c\left(\frac{27}{3} - 0\right) = 9c$

$\Rightarrow 9 \cdot c = 1 \quad \Rightarrow \quad c = \frac{1}{9}$

b) $\mathbb{P}(1 \le X \le 2) = \int_1^2 f_X(x)\,dx = \int_1^2 \frac{x^2}{9}\,dx =$

$= \frac{8-1}{27} = \frac{7}{27}$

$\mathbb{P}(X = 1.5) = 0$

c) $F_X(x) = \mathbb{P}(X \le x) = \int_0^x f_X(t)\,dt = \int_0^x \frac{t^2}{9}\,dt = \frac{x^3}{27}$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^3/27 & 0 \le x \le 3 \\ 1 & x > 3 \end{cases}$$

d) $\mathbb{E}[X] = \int_0^3 x \cdot f_X(x)\,dx = \int_0^3 x \cdot \frac{x^2}{9}\,dx = \int_0^3 \frac{x^3}{9}\,dx = \frac{x^4}{9 \cdot 4}\Big|_0^3 =$

$= \frac{3^4}{36} = \frac{9}{4}$

$\text{var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

$$\mathbb{E}[X^2] = \int_0^3 x^2 \cdot f_X(x)\, dx = \int_0^3 x^2 \cdot \frac{x^2}{9}\, dx = \int_0^3 \frac{x^4}{9}\, dx = \frac{x^5}{5 \cdot 9}\Big|_0^3$$

$$= \ldots = \frac{27}{5}$$

$$\text{var}(X) = \frac{27}{5} - \left(\frac{9}{4}\right)^2 = \frac{27}{80}$$